

Bootstrap 5

Complete Frontend Framework Notes

Grid · Components · Utilities · Flexbox · Cards · Forms · Navbar · Projects

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Module 1 — Introduction to Bootstrap 5

1.1 What is Bootstrap?

Bootstrap is the world's most popular **open-source CSS framework**. It provides ready-made, responsive components — grids, buttons, forms, navbars, modals — so you build professional pages faster with less custom CSS. Version 5 (current) removed jQuery dependency.

Feature	Without Bootstrap	With Bootstrap
Grid layout	Write Flexbox/Grid CSS from scratch	Just add row and col classes
Navbar	50+ lines CSS	5 lines HTML with classes
Responsive	Write media queries	Built in — col-md-6 etc.
Buttons	Custom CSS for each button style	btn btn-primary etc.
Forms	Custom styling for every input	Form-control class
Browser support	Must test and fix manually	Tested across all browsers

1.2 Setting Up Bootstrap — CDN (Quickest)

Add these two lines to your HTML file — no download needed.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My Bootstrap Page</title>

  <!-- Bootstrap 5 CSS – paste in <head> -->
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
        rel="stylesheet">
</head>
<body>

  <h1 class="text-primary text-center mt-5">Hello Bootstrap!</h1>
  <p class="lead text-center">ForkInfosystems, Karad</p>

  <!-- Bootstrap JS Bundle (includes Popper) – paste before </body> -->
  <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.js">
  </script>
</body>
</html>
```

Note: The viewport meta tag is REQUIRED for Bootstrap's responsive grid to work. The JS bundle is only needed for interactive components: modal, tooltip, dropdown, carousel etc.

1.3 How Bootstrap Classes Work

Bootstrap works by applying **utility classes** directly to HTML elements. No custom CSS needed for most layouts.

```
<!-- Without Bootstrap -->
<button style="background:#0d6efd;color:white;padding:8px 16px;
            border:none;border-radius:4px;">Click Me</button>

<!-- With Bootstrap – same result, much cleaner -->
<button class="btn btn-primary">Click Me</button>

<!-- Text alignment -->
<p class="text-center">Centred text</p>
<p class="text-end">Right-aligned</p>

<!-- Colours -->
<p class="text-primary">Blue text</p>
<p class="text-danger">Red text</p>
<div class="bg-warning p-3">Yellow background with padding</div>

<!-- Margin & Padding – m=margin, p=padding, t/b/s/e/x/y = side -->
<div class="mt-4 mb-2 px-3">Spaced div</div>
<!-- mt-4 = margin-top: 1.5rem, px-3 = padding-left+right: 1rem -->
```

1.4 Bootstrap 5 File Structure (Download)

```
<!-- If downloading Bootstrap (not CDN), file structure: -->
my-project/
├── index.html
├── css/
│   ├── style.css          (your custom CSS – always AFTER bootstrap.min.css)
├── js/
│   ├── script.js          (your custom JS)
├── bootstrap-5.3.3/
│   ├── css/
│   │   ├── bootstrap.min.css  (include this)
│   │   ├── bootstrap.min.css.map
│   │   └── js/
│   │       ├── bootstrap.bundle.min.js  (include this – has Popper built in)
│   │       └── bootstrap.min.js        (without Popper – use only if needed)
└──
```



```
<!-- Link in HTML -->
<link rel="stylesheet" href="bootstrap-5.3.3/css/bootstrap.min.css">
<link rel="stylesheet" href="css/style.css" <!-- your CSS after -->
<script src="bootstrap-5.3.3/js/bootstrap.bundle.min.js"></script>
```

Module 2 — Bootstrap Grid System

2.1 How the Grid Works

Bootstrap uses a **12-column responsive grid**. Every layout needs three layers: **container** → **row** → **col**. Columns automatically stack on small screens.

	Class prefix	Screen width	Container max-width
s)	col-	< 576px	
	col-sm-	≥ 576px	
	col-md-	≥ 768px	
	col-lg-	≥ 992px	
	col-xl-	≥ 1200px	
	col-xxl-	≥ 1400px	

2.2 Container Types

```
<!-- container — fixed max-width, centred -->
<div class="container">
  Content here (max-width changes at each breakpoint)
</div>

<!-- container-fluid — always 100% width -->
<div class="container-fluid">
  Full-width content
</div>

<!-- container-md — fluid until md breakpoint, then fixed -->
<div class="container-md">
  Fluid on mobile, fixed from 768px upward
</div>
```

2.3 Basic Grid Examples

```
<!-- Equal columns – col divides 12 equally -->
<div class="container">
  <div class="row">
    <div class="col bg-primary text-white p-3">Column 1</div>
    <div class="col bg-secondary text-white p-3">Column 2</div>
    <div class="col bg-success text-white p-3">Column 3</div>
  </div>
</div>

<!-- Specific sizes – must add up to 12 (or less) -->
<div class="row">
  <div class="col-8 bg-info p-3">8 columns wide</div>
  <div class="col-4 bg-warning p-3">4 columns wide</div>
</div>

<!-- Responsive: full on mobile, half on tablet, third on desktop -->
<div class="row">
  <div class="col-12 col-md-6 col-lg-4 p-2 border">Card 1</div>
  <div class="col-12 col-md-6 col-lg-4 p-2 border">Card 2</div>
  <div class="col-12 col-md-6 col-lg-4 p-2 border">Card 3</div>
</div>

<!-- Offset – push column to the right -->
<div class="row">
  <div class="col-md-4 offset-md-4 bg-light p-3 text-center">
    Centred using offset
  </div>
</div>

<!-- Nested rows -->
<div class="row">
  <div class="col-md-8">
    Main content
    <div class="row mt-3">
      <div class="col-6 bg-light p-2">Sub col 1</div>
      <div class="col-6 bg-light p-2">Sub col 2</div>
    </div>
  </div>
  <div class="col-md-4 bg-secondary text-white p-3">Sidebar</div>
</div>
```

2.4 Row & Column Alignment

```
<!-- Vertical alignment of columns in a row -->
<div class="row align-items-start">    <!-- top -->
<div class="row align-items-center">    <!-- middle -->
<div class="row align-items-end">        <!-- bottom -->

<!-- Horizontal alignment -->
<div class="row justify-content-start">
<div class="row justify-content-center">
<div class="row justify-content-end">
<div class="row justify-content-between">
<div class="row justify-content-around">

<!-- No gutters (remove gap between columns) -->
<div class="row g-0">...</div>

<!-- Custom gutter sizes (g-1 to g-5) -->
<div class="row g-3">...</div>        <!-- gap between columns -->
<div class="row gx-4 gy-2">...</div> <!-- gx=horizontal, gy=vertical -->

<!-- Column self-alignment -->
<div class="col align-self-start">Top</div>
<div class="col align-self-center">Middle</div>
<div class="col align-self-end">Bottom</div>

<!-- Auto width – column takes only the space it needs -->
<div class="col-auto">Only as wide as content</div>
```

Module 3 — Typography & Color Utilities

3.1 Typography Classes

Class	Element/Effect	Description
<code>display-1 to 6</code>	Heading	Larger than h1-h6. Display headings.
<code>h1 to h6</code>	Any element	Makes any element look like that heading.
<code>lead</code>	p	Larger, lighter paragraph — intro text.
<code>fw-bold</code>	Any	font-weight: bold (700)
<code>fw-semibold</code>	Any	font-weight: 600
<code>fw-normal</code>	Any	font-weight: normal (400)
<code>fw-light</code>	Any	font-weight: 300
<code>fst-italic</code>	Any	font-style: italic
<code>text-decoration-none</code>	a	Removes underline from links
<code>text-uppercase</code>	Any	UPPERCASE text
<code>text-lowercase</code>	Any	lowercase text
<code>text-capitalize</code>	Any	Capitalise Each Word
<code>fs-1 to fs-6</code>	Any	Font size 1 (largest) to 6 (smallest)
<code>lh-1 lh-sm lh-base lh-lg</code>	Any	Line height variants
<code>text-truncate</code>	Block element	Truncate overflow text with ...
<code>text-wrap</code>	Any	Wrap long text
<code>text-nowrap</code>	Any	Prevent text wrapping
<code>font-monospace</code>	Any	Use monospace font (like code)
<code>text-muted</code>	Any	Greyed-out / subdued text colour

3.2 Text Alignment & Color

```
<!-- Text alignment – responsive -->
<p class="text-start">Left (default)</p>
<p class="text-center">Centred</p>
<p class="text-end">Right</p>
<p class="text-md-center">Centre only on md and up</p>

<!-- Text colours -->
<p class="text-primary">Primary (blue)</p>
<p class="text-secondary">Secondary (grey)</p>
<p class="text-success">Success (green)</p>
<p class="text-danger">Danger (red)</p>
<p class="text-warning">Warning (yellow)</p>
<p class="text-info">Info (cyan)</p>
<p class="text-light bg-dark">Light (white)</p>
<p class="text-dark">Dark</p>
<p class="text-muted">Muted (grey)</p>
<p class="text-white bg-dark">White</p>

<!-- Background colours -->
<div class="bg-primary text-white p-3">Primary bg</div>
<div class="bg-success text-white p-3">Success bg</div>
<div class="bg-danger text-white p-3">Danger bg</div>
<div class="bg-warning p-3">Warning bg</div>
<div class="bg-light p-3">Light bg</div>
<div class="bg-dark text-white p-3">Dark bg</div>

<!-- Opacity variants (BS 5.1+) -->
<div class="bg-primary bg-opacity-25 p-3">25% opacity primary</div>
<div class="bg-primary bg-opacity-50 p-3">50% opacity</div>
```

3.3 Blockquote, Lists & Code

```
<!-- Blockquote -->
<figure>
  <blockquote class="blockquote">
    <p>Learning by doing is the only way to master web development.</p>
  </blockquote>
  <figcaption class="blockquote-footer">
    Ranjiit Sir, <cite>ForkInfosystems</cite>
  </figcaption>
</figure>

<!-- Unstyled list -->
<ul class="list-unstyled">
  <li>No bullets, no padding</li>
  <li>Clean list</li>
</ul>

<!-- Inline list -->
<ul class="list-inline">
  <li class="list-inline-item">HTML</li>
  <li class="list-inline-item">CSS</li>
  <li class="list-inline-item">Bootstrap</li>
</ul>

<!-- Inline code -->
<p>Use the <code>display-1</code> class for large headings.</p>

<!-- Code block -->
<pre><code>
let x = 10;
console.log(x);
</code></pre>

<!-- Keyboard input -->
<p>Press <kbd>Ctrl</kbd> + <kbd>S</kbd> to save.</p>
```

Module 4 — Spacing, Sizing & Display Utilities

4.1 Spacing — Margin & Padding

Bootstrap spacing classes follow the pattern: **{property}{sides}-{breakpoint}-{size}**

Class Pattern	Meaning
m = margin	p = padding
t = top	b = bottom
s = start (left)	e = end (right)
x = left + right	y = top + bottom
(blank) = all sides	
0 = 0px	1 = 0.25rem (4px)
2 = 0.5rem (8px)	3 = 1rem (16px)
4 = 1.5rem (24px)	5 = 3rem (48px)
auto = auto	Used for centering: mx-auto

```

<!-- Margin examples -->
<div class="m-3">All sides 1rem</div>
<div class="mt-4">Top only 1.5rem</div>
<div class="mb-2">Bottom only 0.5rem</div>
<div class="mx-auto" style="width:300px">Horizontally centred</div>
<div class="my-5">Top & bottom 3rem</div>
<div class="ms-3">Start (left) 1rem</div>

<!-- Padding examples -->
<div class="p-4">All sides 1.5rem padding</div>
<div class="px-5 py-3">Horizontal 3rem, vertical 1rem</div>
<div class="pt-2 pb-4">Top 0.5rem, bottom 1.5rem</div>

<!-- Responsive spacing -->
<div class="mt-2 mt-md-5">Small margin on mobile, large on desktop</div>

<!-- Negative margin (BS5 – use with care) -->
<div class="mt-n3">Negative top margin</div>

<!-- Gap (for flexbox/grid containers) -->
<div class="d-flex gap-3">
  <div>Item 1</div>
  <div>Item 2</div>
</div>

```

4.2 Sizing Utilities

```
<!-- Width -->
<div class="w-25 bg-primary">25% wide</div>
<div class="w-50 bg-secondary">50% wide</div>
<div class="w-75 bg-info">75% wide</div>
<div class="w-100 bg-success">100% wide</div>
<div class="w-auto bg-warning">Auto width</div>
<div class="mw-100">Max width 100%</div>

<!-- Height -->
<div class="h-25">25% height</div>
<div class="h-50">50% height</div>
<div class="h-100">100% height</div>
<div class="mh-100">Max height 100%</div>

<!-- Viewport sizing -->
<div class="vw-100">100 viewport width</div>
<div class="vh-100">100 viewport height</div>
<div class="min-vw-100">Min 100vw</div>
<div class="min-vh-100">Min 100vh – full page height</div>
```

4.3 Display & Visibility

```
<!-- Display utilities: d-{value} -->
<div class="d-none">Hidden (display:none)</div>
<div class="d-block">Block</div>
<div class="d-inline">Inline</div>
<div class="d-inline-block">Inline-block</div>
<div class="d-flex">Flex container</div>
<div class="d-grid">Grid container</div>

<!-- Responsive display – show/hide at breakpoints -->
<div class="d-none d-md-block">Hidden on mobile, visible on md+</div>
<div class="d-block d-md-none">Visible on mobile, hidden on md+</div>
<div class="d-none d-lg-flex">Flex only on large screens</div>

<!-- Visibility (takes space but not visible) -->
<div class="visible">Visible (default)</div>
<div class="invisible">Invisible but takes space</div>

<!-- Overflow -->
<div class="overflow-auto" style="height:100px">Scrollable content</div>
<div class="overflow-hidden">Clipped content</div>
<div class="overflow-visible">Visible overflow</div>
<div class="overflow-scroll">Always scrollbar</div>

<!-- Position -->
<div class="position-relative">Relative</div>
<div class="position-absolute top-0 start-0">Top-left corner</div>
<div class="position-fixed bottom-0 end-0">Fixed bottom-right</div>
<div class="position-sticky top-0">Sticky top</div>
```

Module 5 — Flexbox Utilities

5.1 Flex Container Classes

Class	CSS Property	Options
<code>d-flex</code>	<code>display: flex</code>	Makes element a flex container
<code>d-inline-flex</code>	<code>display: inline-flex</code>	Inline flex container
<code>flex-row</code>	<code>flex-direction: row</code>	default — left to right
<code>flex-column</code>	<code>flex-direction: column</code>	top to bottom
<code>flex-row-reverse</code>	<code>flex-direction: row-reverse</code>	right to left
<code>flex-column-reverse</code>	<code>flex-direction: column-reverse</code>	bottom to top
<code>flex-wrap</code>	<code>flex-wrap: wrap</code>	Items wrap to next line
<code>flex-nowrap</code>	<code>flex-wrap: nowrap</code>	default — no wrapping
<code>flex-wrap-reverse</code>	<code>flex-wrap: wrap-reverse</code>	Wrap in reverse
<code>justify-content-start</code>	<code>justify-content: flex-start</code>	Left (default)
<code>justify-content-end</code>	<code>justify-content: flex-end</code>	Right
<code>justify-content-center</code>	<code>justify-content: center</code>	Centre
<code>justify-content-between</code>	<code>justify-content: space-between</code>	Space between
<code>justify-content-around</code>	<code>justify-content: space-around</code>	Space around
<code>justify-content-evenly</code>	<code>justify-content: space-evenly</code>	Equal spacing
<code>align-items-start</code>	<code>align-items: flex-start</code>	Top
<code>align-items-center</code>	<code>align-items: center</code>	Middle
<code>align-items-end</code>	<code>align-items: flex-end</code>	Bottom
<code>align-items-stretch</code>	<code>align-items: stretch</code>	Stretch to fill
<code>gap-{0-5}</code>	<code>gap: ...</code>	Space between flex items

5.2 Flex Item Classes

```
<!-- Flex item utilities -->
<div class="d-flex">
  <div class="flex-fill">Takes equal share of remaining space</div>
  <div class="flex-grow-1">Grows to fill space</div>
  <div class="flex-grow-0">Does not grow</div>
  <div class="flex-shrink-1">Shrinks if needed</div>
  <div class="flex-shrink-0">Never shrinks</div>
  <div class="order-1">Displayed first</div>
  <div class="order-2">Displayed second</div>
  <div class="order-last">Always last</div>
  <div class="order-first">Always first</div>
  <div class="align-self-start">Align to top</div>
  <div class="align-self-center">Align to centre</div>
  <div class="align-self-end">Align to bottom</div>
  <div class="ms-auto">Push to right (auto margin)</div>
</div>
```

5.3 Practical Flexbox Patterns

```
<!-- 1. Horizontal navbar with space-between -->
<div class="d-flex justify-content-between align-items-center p-3 bg-dark">
  <span class="text-white fw-bold">ForkInfosystems</span>
  <div class="d-flex gap-3">
    <a href="#" class="text-white text-decoration-none">Home</a>
    <a href="#" class="text-white text-decoration-none">Courses</a>
    <a href="#" class="text-white text-decoration-none">Contact</a>
  </div>
</div>

<!-- 2. Centre anything (vertically + horizontally) -->
<div class="d-flex justify-content-center align-items-center vh-100">
  <div class="text-center">
    <h1>Centred Content</h1>
    <p>Perfectly in the middle</p>
  </div>
</div>

<!-- 3. Card row that wraps -->
<div class="d-flex flex-wrap gap-3">
  <div class="card" style="width:250px">Card 1</div>
  <div class="card" style="width:250px">Card 2</div>
  <div class="card" style="width:250px">Card 3</div>
</div>

<!-- 4. Sidebar + main layout -->
<div class="d-flex">
  <div class="bg-dark text-white p-3" style="width:240px; min-height:100vh">
    Sidebar
  </div>
  <div class="flex-fill p-4">
    Main Content
  </div>
</div>

<!-- 5. Push button to the right inside a card footer -->
<div class="card-footer d-flex justify-content-end gap-2">
  <button class="btn btn-outline-secondary">Cancel</button>
  <button class="btn btn-primary">Save</button>
</div>
```

Module 6 — Buttons, Badges & Alerts

6.1 Buttons — All Styles

```
<!-- Solid buttons -->
<button class="btn btn-primary">Primary</button>
<button class="btn btn-secondary">Secondary</button>
<button class="btn btn-success">Success</button>
<button class="btn btn-danger">Danger</button>
<button class="btn btn-warning">Warning</button>
<button class="btn btn-info">Info</button>
<button class="btn btn-light">Light</button>
<button class="btn btn-dark">Dark</button>

<!-- Outline buttons -->
<button class="btn btn-outline-primary">Outline Primary</button>
<button class="btn btn-outline-danger">Outline Danger</button>

<!-- Button sizes -->
<button class="btn btn-primary btn-lg">Large</button>
<button class="btn btn-primary">Default</button>
<button class="btn btn-primary btn-sm">Small</button>

<!-- Full-width button -->
<button class="btn btn-success w-100">Full Width</button>

<!-- Disabled state -->
<button class="btn btn-primary" disabled>Disabled</button>

<!-- Loading state – combine with spinner -->
<button class="btn btn-primary" disabled>
  <span class="spinner-border spinner-border-sm me-2"></span>
  Loading...
</button>

<!-- Button as link -->
<a href="#" class="btn btn-primary">Link Button</a>

<!-- Toggle button -->
<button class="btn btn-outline-primary" data-bs-toggle="button">
  Toggle Me
</button>
```

6.2 Button Groups

```
<!-- Button group -->
<div class="btn-group">
  <button class="btn btn-primary">Left</button>
  <button class="btn btn-primary">Middle</button>
  <button class="btn btn-primary">Right</button>
</div>

<!-- Vertical button group -->
<div class="btn-group-vertical">
  <button class="btn btn-outline-primary">Top</button>
  <button class="btn btn-outline-primary">Middle</button>
  <button class="btn btn-outline-primary">Bottom</button>
</div>

<!-- Radio button group (like tabs) -->
<div class="btn-group" role="group">
  <input type="radio" class="btn-check" name="view" id="list" checked>
  <label class="btn btn-outline-primary" for="list">List View</label>
  <input type="radio" class="btn-check" name="view" id="grid">
  <label class="btn btn-outline-primary" for="grid">Grid View</label>
</div>
```

6.3 Badges

```
<!-- Inline badges -->
<h5>Notifications <span class="badge bg-danger">4</span></h5>
<button class="btn btn-primary">
  Cart <span class="badge bg-warning text-dark ms-1">3</span>
</button>

<!-- Badge colours -->
<span class="badge bg-primary">Primary</span>
<span class="badge bg-success">Success</span>
<span class="badge bg-danger">Danger</span>
<span class="badge bg-warning text-dark">Warning</span>
<span class="badge bg-info text-dark">Info</span>
<span class="badge bg-dark">Dark</span>

<!-- Pill badge (rounded) -->
<span class="badge rounded-pill bg-primary">New</span>
<span class="badge rounded-pill bg-danger">99+</span>
```

6.4 Alerts

```
<!-- Alert variants -->
<div class="alert alert-primary">A simple primary alert</div>
<div class="alert alert-success">Successfully saved!</div>
<div class="alert alert-danger">Something went wrong!</div>
<div class="alert alert-warning">Please check your input.</div>
<div class="alert alert-info">Your session expires in 5 minutes.</div>

<!-- Dismissible alert -->
<div class="alert alert-warning alert-dismissible fade show">
  <strong>Warning!</strong> Your subscription expires tomorrow.
  <button type="button" class="btn-close" data-bs-dismiss="alert"></button>
</div>

<!-- Alert with icon and link -->
<div class="alert alert-success d-flex align-items-center gap-2">
  <span>&#10003;</span>
  <div>
    Profile updated. <a href="#" class="alert-link">View changes</a>
  </div>
</div>
```

Module 7 — Navigation Components

7.1 Navbar — Responsive Navigation Bar

```
<nav class="navbar navbar-expand-lg navbar-dark bg-primary">
  <div class="container">

    <!-- Brand / Logo -->
    <a class="navbar-brand fw-bold" href="#">
      ForkInfo<span class="text-warning">systems</span>
    </a>

    <!-- Hamburger toggle for mobile -->
    <button class="navbar-toggler" type="button"
      data-bs-toggle="collapse" data-bs-target="#navMenu">
      <span class="navbar-toggler-icon"></span>
    </button>

    <!-- Nav links — collapse on mobile -->
    <div class="collapse navbar-collapse" id="navMenu">
      <ul class="navbar-nav me-auto mb-2 mb-lg-0">
        <li class="nav-item">
          <a class="nav-link active" href="#">Home</a>
        </li>
        <li class="nav-item">
          <a class="nav-link" href="#">Courses</a>
        </li>
        <!-- Dropdown -->
        <li class="nav-item dropdown">
          <a class="nav-link dropdown-toggle" href="#"
            data-bs-toggle="dropdown">More</a>
          <ul class="dropdown-menu">
            <li><a class="dropdown-item" href="#">About Us</a></li>
            <li><a class="dropdown-item" href="#">Gallery</a></li>
            <li><hr class="dropdown-divider"></li>
            <li><a class="dropdown-item" href="#">Contact</a></li>
          </ul>
        </li>
      </ul>
      <!-- Search form in navbar -->
      <form class="d-flex gap-2">
        <input class="form-control" type="search" placeholder="Search...">
        <button class="btn btn-warning" type="submit">Search</button>
      </form>
    </div>

  </div>
</nav>
```

Tip: navbar-expand-lg means the navbar is expanded (horizontal) on large screens and collapses to a hamburger menu on smaller screens. Change lg to md or sm to control when it collapses.

7.2 Nav Tabs & Pills

```
<!-- Tab navigation -->
<ul class="nav nav-tabs">
  <li class="nav-item">
    <a class="nav-link active" href="#">HTML</a>
  </li>
  <li class="nav-item">
    <a class="nav-link" href="#">CSS</a>
  </li>
  <li class="nav-item">
    <a class="nav-link disabled">JavaScript</a>
  </li>
</ul>

<!-- Pill navigation -->
<ul class="nav nav-pills">
  <li class="nav-item">
    <a class="nav-link active" href="#">Home</a>
  </li>
  <li class="nav-item">
    <a class="nav-link" href="#">Profile</a>
  </li>
</ul>

<!-- Fill – items take equal width -->
<ul class="nav nav-pills nav-fill">
  <li class="nav-item"><a class="nav-link active" href="#">Active</a></li>
  <li class="nav-item"><a class="nav-link" href="#">Link</a></li>
  <li class="nav-item"><a class="nav-link" href="#">Link</a></li>
</ul>

<!-- Justified – items fill full width equally -->
<ul class="nav nav-tabs nav-justified">
  <li class="nav-item"><a class="nav-link active" href="#">Tab 1</a></li>
  <li class="nav-item"><a class="nav-link" href="#">Tab 2</a></li>
  <li class="nav-item"><a class="nav-link" href="#">Tab 3</a></li>
</ul>
```

7.3 Breadcrumb & Pagination

```
<!-- Breadcrumb -->
<nav aria-label="breadcrumb">
  <ol class="breadcrumb">
    <li class="breadcrumb-item"><a href="#">Home</a></li>
    <li class="breadcrumb-item"><a href="#">Courses</a></li>
    <li class="breadcrumb-item active">Web Development</li>
  </ol>
</nav>

<!-- Pagination -->
<nav>
  <ul class="pagination">
    <li class="page-item disabled">
      <a class="page-link" href="#">&laquo; Previous</a>
    </li>
    <li class="page-item"><a class="page-link" href="#">1</a></li>
    <li class="page-item active"><a class="page-link" href="#">2</a></li>
    <li class="page-item"><a class="page-link" href="#">3</a></li>
    <li class="page-item">
      <a class="page-link" href="#">Next &raquo;</a>
    </li>
  </ul>
</nav>

<!-- Pagination sizes & alignment -->
<ul class="pagination pagination-lg justify-content-center">
<ul class="pagination pagination-sm justify-content-end">
```

Module 8 — Cards, Images & Figures

8.1 Bootstrap Cards

Cards are flexible content containers. They are the most-used Bootstrap component for displaying content in boxes.

```

<!-- Basic card structure -->
<div class="card" style="width: 320px;">
  
  <div class="card-body">
    <h5 class="card-title">Web Development</h5>
    <h6 class="card-subtitle mb-2 text-muted">HTML · CSS · JavaScript</h6>
    <p class="card-text">
      Learn to build professional websites from scratch.
      Includes real projects and job preparation.
    </p>
    <a href="#" class="btn btn-primary">Enroll Now</a>
    <a href="#" class="btn btn-outline-secondary ms-2">Learn More</a>
  </div>
  <div class="card-footer text-muted d-flex justify-content-between">
    <small>Updated Jan 2025</small>
    <small>&#9733;&#9733;&#9733;&#9733;&#9733; 4.9</small>
  </div>
</div>

<!-- Card with header -->
<div class="card">
  <div class="card-header bg-primary text-white">
    <strong>Student Results</strong>
  </div>
  <div class="card-body">
    <p class="card-text">Content goes here...</p>
  </div>
</div>

<!-- Horizontal card -->
<div class="card mb-3">
  <div class="row g-0">
    <div class="col-md-4">
      
    </div>
    <div class="col-md-8">
      <div class="card-body">
        <h5 class="card-title">Ranjiit Sir</h5>
        <p class="card-text">Lead Instructor – ForkInfosystems</p>
        <p class="text-muted"><small>Karad, Maharashtra</small></p>
      </div>
    </div>
  </div>
</div>

```

8.2 Card Grid & Card Groups

```
<!-- Responsive card grid - Bootstrap row/col -->
<div class="row row-cols-1 row-cols-md-2 row-cols-lg-3 g-4">
  <div class="col">
    <div class="card h-100">
      <div class="card-body">
        <h5 class="card-title">HTML & CSS</h5>
        <p class="card-text">Build web pages from scratch.</p>
      </div>
      <div class="card-footer">
        <a href="#" class="btn btn-sm btn-primary w-100">Enroll</a>
      </div>
    </div>
  </div>
  <div class="col">
    <div class="card h-100">
      <!-- h-100 makes all cards equal height! -->
      <div class="card-body">
        <h5 class="card-title">JavaScript</h5>
        <p class="card-text">Add interactivity to your pages.</p>
      </div>
      <div class="card-footer">
        <a href="#" class="btn btn-sm btn-primary w-100">Enroll</a>
      </div>
    </div>
  </div>
</div>

<!-- Card colour variants -->
<div class="card text-white bg-primary mb-3">...</div>
<div class="card text-white bg-danger mb-3">...</div>
<div class="card border-success mb-3">
  <div class="card-header text-success">Success border</div>
  <div class="card-body text-success">...</div>
</div>
```

8.3 Images & Responsive Images

```
<!-- Responsive image – shrinks to fit container -->


<!-- Rounded corners -->




<!-- Thumbnail (adds border and padding) -->


<!-- Float and centre -->




<!-- Object fit (Bootstrap 5.3+) -->

<!-- object-fit-cover | object-fit-contain | object-fit-fill -->

<!-- Ratio boxes – maintain aspect ratio -->
<div class="ratio ratio-16x9">
  <iframe src="https://www.youtube.com/embed/VIDEO_ID" allowfullscreen></iframe>
</div>
<div class="ratio ratio-4x3">...</div>
<div class="ratio ratio-1x1">Square box</div>
```

Module 9 — Forms & Inputs

9.1 Bootstrap Form Structure

```
<!-- Basic form structure -->
<form>
  <!-- Text input -->
  <div class="mb-3">
    <label for="name" class="form-label">Full Name</label>
    <input type="text" class="form-control" id="name"
      placeholder="Enter your full name">
    <div class="form-text text-muted">As on your certificate.</div>
  </div>

  <!-- Email -->
  <div class="mb-3">
    <label for="email" class="form-label">Email Address</label>
    <input type="email" class="form-control" id="email" required>
  </div>

  <!-- Password -->
  <div class="mb-3">
    <label for="pass" class="form-label">Password</label>
    <input type="password" class="form-control" id="pass">
  </div>

  <!-- Textarea -->
  <div class="mb-3">
    <label for="msg" class="form-label">Message</label>
    <textarea class="form-control" id="msg" rows="4"
      placeholder="Write your message..."></textarea>
  </div>

  <!-- Select -->
  <div class="mb-3">
    <label for="course" class="form-label">Select Course</label>
    <select class="form-select" id="course">
      <option selected>-- Choose a Course --</option>
      <option>HTML & CSS</option>
      <option>JavaScript</option>
      <option>Bootstrap</option>
    </select>
  </div>

  <!-- Checkbox -->
  <div class="mb-3 form-check">
    <input type="checkbox" class="form-check-input" id="agree">
    <label class="form-check-label" for="agree">
      I agree to the Terms & Conditions
    </label>
  </div>

  <!-- Radio buttons -->
  <div class="mb-3">
    <div class="form-check">
      <input class="form-check-input" type="radio" name="gender"
        id="male" value="male">
```

```
<label class="form-check-label" for="male">Male</label>
</div>
<div class="form-check">
  <input class="form-check-input" type="radio" name="gender"
    id="female" value="female">
  <label class="form-check-label" for="female">Female</label>
</div>
</div>

<!-- Range slider -->
<div class="mb-3">
  <label for="rating" class="form-label">Rating: <span id="rval">5</span></label>
  <input type="range" class="form-range" id="rating" min="1" max="10" value="5"
    oninput="document.getElementById('rval').textContent=this.value">
</div>

<button type="submit" class="btn btn-primary">Submit</button>
<button type="reset" class="btn btn-outline-secondary ms-2">Reset</button>
</form>
```

9.2 Floating Labels, Input Groups & Sizing

```
<!-- Floating labels (label moves up when input is focused) -->
<div class="form-floating mb-3">
  <input type="email" class="form-control" id="floatEmail"
    placeholder="name@example.com">
  <label for="floatEmail">Email address</label>
</div>

<div class="form-floating mb-3">
  <textarea class="form-control" id="floatMsg"
    style="height:100px" placeholder="Leave a message"></textarea>
  <label for="floatMsg">Message</label>
</div>

<!-- Input group – prepend/append text or buttons -->
<div class="input-group mb-3">
  <span class="input-group-text">@</span>
  <input type="text" class="form-control" placeholder="Username">
</div>

<div class="input-group mb-3">
  <input type="text" class="form-control" placeholder="Search...">
  <button class="btn btn-primary">Search</button>
</div>

<div class="input-group mb-3">
  <span class="input-group-text">₹</span>
  <input type="number" class="form-control" placeholder="Amount">
  <span class="input-group-text">.00</span>
</div>

<!-- Form sizes -->
<input class="form-control form-control-lg" placeholder="Large input">
<input class="form-control" placeholder="Default input">
<input class="form-control form-control-sm" placeholder="Small input">

<!-- Inline form -->
<form class="row g-3 align-items-center">
  <div class="col-auto">
    <input type="text" class="form-control" placeholder="Your name">
  </div>
  <div class="col-auto">
    <button class="btn btn-primary">Submit</button>
  </div>
</form>
```

9.3 Form Validation

```
<!-- Bootstrap validation states (add was-validated to form) -->
<form class="needs-validation" novalidate>
  <div class="mb-3">
    <label for="vName" class="form-label">Name *</label>
    <input type="text" class="form-control" id="vName" required>
    <div class="valid-feedback">Looks good!</div>
    <div class="invalid-feedback">Please enter your name.</div>
  </div>

  <div class="mb-3">
    <label for="vEmail" class="form-label">Email *</label>
    <input type="email" class="form-control" id="vEmail" required>
    <div class="valid-feedback">Valid email!</div>
    <div class="invalid-feedback">Please enter a valid email.</div>
  </div>

  <button class="btn btn-primary" type="submit">Submit</button>
</form>

<script>
// Prevent submission and show validation
const form = document.querySelector('.needs-validation');
form.addEventListener('submit', e => {
  if (!form.checkValidity()) {
    e.preventDefault();
    e.stopPropagation();
  }
  form.classList.add('was-validated');
});
</script>
```

Module 10 — Bootstrap Components

10.1 Modal (Popup Dialog)

```
<!-- Trigger button -->
<button class="btn btn-primary" data-bs-toggle="modal"
        data-bs-target="#myModal">
  Open Modal
</button>

<!-- Modal structure -->
<div class="modal fade" id="myModal" tabindex="-1">
  <div class="modal-dialog modal-dialog-centered">
    <!-- modal-dialog-centered = vertically centred -->
    <!-- Add modal-lg or modal-sm for size -->
    <div class="modal-content">
      <div class="modal-header">
        <h5 class="modal-title">Confirm Enrollment</h5>
        <button class="btn-close" data-bs-dismiss="modal"></button>
      </div>
      <div class="modal-body">
        <p>Are you sure you want to enroll in <strong>Web Development</strong>?</p>
        <p class="text-muted">Fee: ₹8377;5,000 | Duration: 3 months</p>
      </div>
      <div class="modal-footer">
        <button class="btn btn-secondary" data-bs-dismiss="modal">Cancel</button>
        <button class="btn btn-primary">Yes, Enroll!</button>
      </div>
    </div>
  </div>
</div>
```

10.2 Accordion (Collapsible Sections)

```
<div class="accordion" id="faqAccordion">

  <div class="accordion-item">
    <h2 class="accordion-header">
      <button class="accordion-button" type="button"
        data-bs-toggle="collapse" data-bs-target="#faq1">
        What is Bootstrap?
      </button>
    </h2>
    <div id="faq1" class="accordion-collapse collapse show"
      data-bs-parent="#faqAccordion">
      <div class="accordion-body">
        Bootstrap is a CSS framework that provides ready-made responsive components.
      </div>
    </div>
  </div>

  <div class="accordion-item">
    <h2 class="accordion-header">
      <button class="accordion-button collapsed" type="button"
        data-bs-toggle="collapse" data-bs-target="#faq2">
        Do I need to know CSS before Bootstrap?
      </button>
    </h2>
    <div id="faq2" class="accordion-collapse collapse"
      data-bs-parent="#faqAccordion">
      <div class="accordion-body">
        Yes – basic CSS knowledge is recommended before using Bootstrap.
      </div>
    </div>
  </div>

</div>
```

10.3 Carousel (Image Slider)

```
<div id="heroCarousel" class="carousel slide" data-bs-ride="carousel">

  <!-- Indicators (dots at bottom) -->
  <div class="carousel-indicators">
    <button data-bs-target="#heroCarousel" data-bs-slide-to="0" class="active"></button>
    <button data-bs-target="#heroCarousel" data-bs-slide-to="1"></button>
    <button data-bs-target="#heroCarousel" data-bs-slide-to="2"></button>
  </div>

  <!-- Slides -->
  <div class="carousel-inner">
    <div class="carousel-item active">
      
      <div class="carousel-caption">
        <h3>Web Development</h3>
        <p>Learn HTML, CSS, Bootstrap & JavaScript</p>
        <a href="#" class="btn btn-primary">Enroll Now</a>
      </div>
    </div>
    <div class="carousel-item">
      
    </div>
  </div>

  <!-- Controls (prev/next arrows) -->
  <button class="carousel-control-prev" data-bs-target="#heroCarousel"
    data-bs-slide="prev">
    <span class="carousel-control-prev-icon"></span>
  </button>
  <button class="carousel-control-next" data-bs-target="#heroCarousel"
    data-bs-slide="next">
    <span class="carousel-control-next-icon"></span>
  </button>
</div>
```

10.4 Spinners, Toast & Progress

```
<!-- Spinners -->
<div class="spinner-border text-primary" role="status">
  <span class="visually-hidden">Loading...</span>
</div>
<div class="spinner-grow text-danger"></div>

<!-- Small spinner in button -->
<button class="btn btn-primary disabled">
  <span class="spinner-border spinner-border-sm me-2"></span>Please wait...
</button>

<!-- Progress bar -->
<div class="progress mb-3" style="height:20px;">
  <div class="progress-bar bg-success" style="width:75%">75%</div>
</div>

<!-- Striped + animated progress -->
<div class="progress mb-3">
  <div class="progress-bar progress-bar-striped progress-bar-animated
    bg-primary" style="width:60%"></div>
</div>

<!-- Multiple bars in one -->
<div class="progress">
  <div class="progress-bar bg-success" style="width:30%">HTML</div>
  <div class="progress-bar bg-info" style="width:25%">CSS</div>
  <div class="progress-bar bg-warning" style="width:20%">JS</div>
</div>

<!-- Toast notification -->
<div class="toast-container position-fixed bottom-0 end-0 p-3">
  <div class="toast show" role="alert">
    <div class="toast-header">
      <strong class="me-auto">ForkInfosystems</strong>
      <small>Just now</small>
      <button class="btn-close" data-bs-dismiss="toast"></button>
    </div>
    <div class="toast-body">
      &#10003; Your registration was successful!
    </div>
  </div>
</div>
</div>
```

Module 11 — Bootstrap JavaScript Plugins

11.1 Data Attributes vs JavaScript API

Bootstrap components can be controlled in two ways: 1) purely with HTML **data-bs-*** attributes (no JS needed), 2) programmatically via JavaScript API.

```
<!-- DATA ATTRIBUTE WAY (no JS code needed) -->
<button data-bs-toggle="modal" data-bs-target="#myModal">Open</button>
<button data-bs-toggle="collapse" data-bs-target="#collapseDiv">Toggle</button>
<button data-bs-toggle="tooltip" data-bs-title="Click to learn more!">Hover</button>

<!-- JAVASCRIPT API WAY -->
<script>
  // Modal
  const modal = new bootstrap.Modal(document.getElementById('myModal'));
  modal.show();
  modal.hide();
  modal.toggle();

  // Collapse
  const collapse = new bootstrap.Collapse(document.getElementById('myCollapse'));
  collapse.toggle();
  collapse.show();
  collapse.hide();

  // Tooltip – must initialise ALL tooltips on page load
  const tooltipEls = document.querySelectorAll('[data-bs-toggle="tooltip"]');
  tooltipEls.forEach(el => new bootstrap.Tooltip(el));

  // Toast
  const toast = new bootstrap.Toast(document.getElementById('myToast'));
  toast.show();
  // Options
  const toast2 = new bootstrap.Toast(el, { delay: 3000, autohide: true });

  // Offcanvas sidebar
  const offcanvas = new bootstrap.Offcanvas(document.getElementById('mySidebar'));
  offcanvas.show();
</script>
```

11.2 Offcanvas Sidebar

```
<!-- Trigger -->
<button class="btn btn-primary" data-bs-toggle="offcanvas"
        data-bs-target="#sidebar">
    &#9776; Open Sidebar
</button>

<!-- Offcanvas panel -->
<div class="offcanvas offcanvas-start" id="sidebar">
    <!-- offcanvas-start=left, offcanvas-end=right,
        offcanvas-top, offcanvas-bottom -->
    <div class="offcanvas-header bg-primary text-white">
        <h5 class="offcanvas-title">ForkInfosystems</h5>
        <button class="btn-close btn-close-white"
            data-bs-dismiss="offcanvas"></button>
    </div>
    <div class="offcanvas-body">
        <ul class="nav flex-column gap-2">
            <li><a href="#" class="btn btn-outline-primary w-100 text-start">
                &#127968; Dashboard</a></li>
            <li><a href="#" class="btn btn-outline-primary w-100 text-start">
                &#128218; Courses</a></li>
            <li><a href="#" class="btn btn-outline-primary w-100 text-start">
                &#128100; Profile</a></li>
            <li><a href="#" class="btn btn-outline-danger w-100 text-start">
                &#128682; Logout</a></li>
        </ul>
    </div>
</div>
```

11.3 Scrollspy

```
<!-- Scrollspy – highlights nav link as you scroll past sections -->
<!-- Step 1: Add data-bs-spy to scrolling container (usually body) -->
<body data-bs-spy="scroll" data-bs-target="#mainNav" data-bs-offset="80">

<!-- Step 2: Navigation with IDs matching section IDs -->
<nav id="mainNav" class="navbar navbar-dark bg-dark sticky-top">
  <div class="container">
    <ul class="navbar-nav d-flex flex-row gap-3">
      <li class="nav-item"><a class="nav-link" href="#home">Home</a></li>
      <li class="nav-item"><a class="nav-link" href="#about">About</a></li>
      <li class="nav-item"><a class="nav-link" href="#courses">Courses</a></li>
      <li class="nav-item"><a class="nav-link" href="#contact">Contact</a></li>
    </ul>
  </div>
</nav>

<!-- Step 3: Sections with matching IDs -->
<section id="home" style="height:100vh;">Home section</section>
<section id="about" style="height:100vh;">About section</section>
<section id="courses" style="height:100vh;">Courses section</section>
<section id="contact" style="height:100vh;">Contact section</section>
```

Module 12 — Responsive Utilities & Customisation

12.1 Responsive Utility Classes

Pattern	Example	Meaning
<code>d-{bp}-{val}</code>	<code>d-none d-md-block</code>	Hidden mobile, block on md+
<code>text-{bp}-{align}</code>	<code>text-start text-md-center</code>	Left mobile, centred on md+
<code>float-{bp}-{val}</code>	<code>float-md-end</code>	Float right on md+ screens
<code>order-{bp}-{n}</code>	<code>order-2 order-md-1</code>	Reorder columns at breakpoint
<code>w-{bp}-{val}</code>	<code>w-100 w-md-50</code>	Width changes at breakpoint
<code>gap-{bp}-{n}</code>	<code>gap-2 gap-lg-4</code>	Gap size changes at breakpoint

12.2 Dark Mode (Bootstrap 5.3+)

```

<!-- Enable dark mode by adding data-bs-theme to html tag -->
<html lang="en" data-bs-theme="dark">

<!-- Or on a specific component -->
<div class="card" data-bs-theme="dark">...</div>
<nav class="navbar" data-bs-theme="dark">...</nav>

<!-- Toggle dark mode with JavaScript -->
<button id="theme-toggle" class="btn btn-outline-secondary">
  &#9790; Toggle Dark Mode
</button>

<script>
  document.getElementById('theme-toggle').addEventListener('click', () => {
    const html = document.documentElement;
    const current = html.getAttribute('data-bs-theme');
    html.setAttribute('data-bs-theme', current === 'dark' ? 'light' : 'dark');
  });
</script>

```

12.3 Customising Bootstrap with CSS Variables

```
/* Bootstrap 5 exposes CSS variables – override in your own stylesheet */
/* Place this AFTER bootstrap.min.css link in <head> */

:root {
  --bs-primary:      #1a237e; /* change primary colour */
  --bs-primary-rgb:  26,35,126;
  --bs-secondary:    #6c757d;
  --bs-success:       #2e7d32;
  --bs-danger:        #b71c1c;
  --bs-font-sans-serif: 'Poppins', sans-serif;
  --bs-border-radius: 8px;
  --bs-border-radius-lg: 12px;
}

/* Override specific component */
.btn-primary {
  background-color: var(--bs-primary);
  border-color:     var(--bs-primary);
}
.btn-primary:hover {
  background-color: #e65100;
  border-color:     #e65100;
}

/* Custom card shadow */
.card {
  box-shadow: 0 4px 15px rgba(0,0,0,0.08);
  border: none;
}

/* Custom navbar height */
.navbar { min-height: 70px; }
```

12.4 Useful Utility Classes Quick Reference

Category	Class Examples	Effect
Rounding	rounded rounded-circle rounded-pill	Border radius
Shadow	shadow-none shadow shadow-sm shadow-lg	Box shadow
Border	border border-0 border-top border-primary	Borders
Opacity	opacity-25 opacity-50 opacity-75 opacity-100	Transparency
Cursor	cursor-pointer (custom CSS needed)	Cursor style
Z-index	z-0 z-1 z-2 z-3	Stacking order
Sticky	sticky-top sticky-bottom	Sticky positioning
Clearfix	clearfix	Clear floats
Stretched link	stretched-link on <a>	Make entire card clickable
Visually hidden	visually-hidden	Hidden but accessible

Module 13 — Complete Landing Page Example

A complete, professional Bootstrap 5 landing page for ForkInfosystems. Save as **index.html** — ready to use as a real website.

```

<!DOCTYPE html>
<html lang="en" data-bs-theme="light">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>ForkInfosystems – IT Training, Karad</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
    rel="stylesheet">
  <style>
    :root {
      --bs-primary: #1a237e;
      --primary: #1a237e;
      --accent: #e65100;
    }
    body { font-family: 'Segoe UI', Arial, sans-serif; }
    .navbar { min-height: 65px; }
    .hero {
      background: linear-gradient(135deg, #1a237e 0%, #283593 55%, #e65100 100%);
      min-height: 90vh;
      display: flex; align-items: center;
    }
    .section-title::after {
      content: '';
      display: block;
      width: 55px; height: 4px;
      background: var(--accent);
      margin: 8px auto 0;
      border-radius: 2px;
    }
    .course-card { transition: transform 0.3s, box-shadow 0.3s; }
    .course-card:hover {
      transform: translateY(-6px);
      box-shadow: 0 12px 30px rgba(0,0,0,0.12) !important;
    }
    .stat-box { border-left: 4px solid var(--accent); }
    footer { background: #1a1a2e; }
  </style>
</head>
<body>

<!-- ■■■ NAVBAR ■■■ -->
<nav class="navbar navbar-expand-lg navbar-dark bg-primary sticky-top shadow">
  <div class="container">
    <a class="navbar-brand fw-bold fs-4" href="#">
      Fork<span style="color:#ffc107">Info</span>systems
    </a>
    <button class="navbar-toggler" type="button"
      data-bs-toggle="collapse" data-bs-target="#nav">
      <span class="navbar-toggler-icon"></span>
    </button>
    <div class="collapse navbar-collapse" id="nav">
      <ul class="navbar-nav mx-auto gap-1">
        <li class="nav-item"><a class="nav-link px-3" href="#home">Home</a></li>

```

```

        <li class="nav-item"><a class="nav-link px-3" href="#courses">Courses</a></li>
        <li class="nav-item"><a class="nav-link px-3" href="#about">About</a></li>
        <li class="nav-item"><a class="nav-link px-3" href="#contact">Contact</a></li>
    </ul>
    <a href="#contact" class="btn btn-warning fw-bold px-4">Enroll Now</a>
</div>
</div>
</nav>

<!-- ■■■ HERO ■■■ -->
<section class="hero text-white" id="home">
    <div class="container py-5">
        <div class="row align-items-center g-5">
            <div class="col-lg-7">
                <span class="badge bg-warning text-dark mb-3 fs-6">
                    &#127881; New Batch Starting Soon
                </span>
                <h1 class="display-4 fw-bold mb-3">
                    Build Your Career in<br>
                    <span style="color:#ffc107">Web Development</span>
                </h1>
                <p class="lead mb-4 opacity-90">
                    Industry-ready courses in HTML, CSS, Bootstrap, JavaScript & more.
                    Hands-on training by experienced professionals in Karad.
                </p>
                <div class="d-flex gap-3 flex-wrap">
                    <a href="#courses" class="btn btn-warning btn-lg fw-bold px-4">
                        View Courses &#8594;
                    </a>
                    <a href="#contact" class="btn btn-outline-light btn-lg px-4">
                        Contact Us
                    </a>
                </div>
                <div class="row mt-5 g-3">
                    <div class="col-4 stat-box ps-3">
                        <div class="fs-3 fw-bold text-warning">500+</div>
                        <div class="small opacity-75">Students Trained</div>
                    </div>
                    <div class="col-4 stat-box ps-3">
                        <div class="fs-3 fw-bold text-warning">10+</div>
                        <div class="small opacity-75">Courses Offered</div>
                    </div>
                    <div class="col-4 stat-box ps-3">
                        <div class="fs-3 fw-bold text-warning">95%</div>
                        <div class="small opacity-75">Placement Rate</div>
                    </div>
                </div>
                <div class="col-lg-5 text-center d-none d-lg-block">
                    <div class="display-1">&#128187;</div>
                    <div class="mt-3 d-flex justify-content-center gap-2 flex-wrap">
                        <span class="badge bg-light text-dark fs-6 p-2">HTML</span>

```

```

        <span class="badge bg-light text-dark fs-6 p-2">CSS</span>
        <span class="badge bg-light text-dark fs-6 p-2">Bootstrap</span>
        <span class="badge bg-light text-dark fs-6 p-2">JavaScript</span>
        <span class="badge bg-light text-dark fs-6 p-2">SQL</span>
    </div>
</div>
</div>
</div>
</section>

<!-- ■■ COURSES ■■ -->
<section class="py-5 bg-light" id="courses">
    <div class="container">
        <h2 class="text-center fw-bold section-title mb-5">Our Courses</h2>
        <div class="row g-4">

            <div class="col-md-6 col-lg-4">
                <div class="card h-100 border-0 shadow-sm course-card">
                    <div class="card-body p-4">
                        <div class="fs-1 mb-3">127760</div>
                        <h5 class="fw-bold">HTML & CSS</h5>
                        <p class="text-muted">Build beautiful, responsive web pages from scratch.</p>
                        <ul class="list-unstyled text-muted small">
                            <li>10003; Semantic HTML5</li>
                            <li>10003; Flexbox & Grid</li>
                            <li>10003; Responsive Design</li>
                        </ul>
                    </div>
                    <div class="card-footer bg-white border-0 p-4 pt-0">
                        <div class="d-flex justify-content-between align-items-center">
                            <span class="fw-bold text-primary fs-5">8377;3,500</span>
                            <a href="#contact" class="btn btn-primary btn-sm px-3">Enroll</a>
                        </div>
                    </div>
                </div>
            </div>

            <div class="col-md-6 col-lg-4">
                <div class="card h-100 border-0 shadow-sm course-card">
                    <div class="card-body p-4">
                        <div class="fs-1 mb-3">128187</div>
                        <h5 class="fw-bold">JavaScript</h5>
                        <p class="text-muted">Add interactivity and build dynamic web applications.</p>
                        <ul class="list-unstyled text-muted small">
                            <li>10003; ES6+ Modern JS</li>
                            <li>10003; DOM Manipulation</li>
                            <li>10003; APIs & Fetch</li>
                        </ul>
                    </div>
                    <div class="card-footer bg-white border-0 p-4 pt-0">
                        <div class="d-flex justify-content-between align-items-center">

```

```

        <span class="fw-bold text-primary fs-5">₹8377;4,500</span>
        <a href="#contact" class="btn btn-primary btn-sm px-3">Enroll</a>
      </div>
    </div>
  </div>
</div>

<div class="col-md-6 col-lg-4">
  <div class="card h-100 border-0 shadow-sm course-card border-primary">
    <div class="card-header bg-primary text-white text-center py-2">
      <small class="fw-bold">₹128293; MOST POPULAR</small>
    </div>
    <div class="card-body p-4">
      <div class="fs-1 mb-3">₹9889</div>
      <h5 class="fw-bold">Full Stack Web Dev</h5>
      <p class="text-muted">Complete course: HTML + CSS + Bootstrap + JS + SQL.</p>
      <ul class="list-unstyled text-muted small">
        <li>₹10003; All Web Technologies</li>
        <li>₹10003; Real Projects</li>
        <li>₹10003; Job Assistance</li>
      </ul>
    </div>
    <div class="card-footer bg-white border-0 p-4 pt-0">
      <div class="d-flex justify-content-between align-items-center">
        <span class="fw-bold text-primary fs-5">₹8377;10,000</span>
        <a href="#contact" class="btn btn-warning btn-sm px-3 fw-bold">Enroll</a>
      </div>
    </div>
  </div>
</div>
</section>

<!-- ■■ ABOUT ■■ -->
<section class="py-5" id="about">
  <div class="container">
    <div class="row align-items-center g-5">
      <div class="col-lg-5 text-center">
        <div class="display-1 mb-3">₹128104;₹8205;₹127979</div>
        <h4 class="fw-bold">Ranjiit Sir</h4>
        <p class="text-muted">Lead Instructor<br>ForkInfosystems, Karad</p>
        <div class="d-flex justify-content-center gap-2 mt-3">
          <span class="badge bg-primary p-2">Web Development</span>
          <span class="badge bg-success p-2">SQL / Database</span>
          <span class="badge bg-warning text-dark p-2">Programming</span>
        </div>
      </div>
      <div class="col-lg-7">
        <h2 class="fw-bold section-title mb-4">Why ForkInfosystems?</h2>
        <div class="row g-3">

```

```

        <div class="col-sm-6">
            <div class="card border-0 bg-light p-3 h-100">
                <div class="fs-2 mb-2">&#127919;</div>
                <h6 class="fw-bold">Job-Ready Curriculum</h6>
                <p class="text-muted small mb-0">Industry-aligned content for real placement
s.</p>
            </div>
        </div>
        <div class="col-sm-6">
            <div class="card border-0 bg-light p-3 h-100">
                <div class="fs-2 mb-2">&#128187;</div>
                <h6 class="fw-bold">100% Practical</h6>
                <p class="text-muted small mb-0">Build real projects, not just theory exerci
ses.</p>
            </div>
        </div>
        <div class="col-sm-6">
            <div class="card border-0 bg-light p-3 h-100">
                <div class="fs-2 mb-2">&#128218;</div>
                <h6 class="fw-bold">Study Material</h6>
                <p class="text-muted small mb-0">Professional PDF notes for every module.</p>
            </div>
        </div>
        <div class="col-sm-6">
            <div class="card border-0 bg-light p-3 h-100">
                <div class="fs-2 mb-2">&#129351;</div>
                <h6 class="fw-bold">Certificate</h6>
                <p class="text-muted small mb-0">Industry-recognised course certificate.</p>
            </div>
        </div>
    </div>
</div>
</div>
</div>
</section>

<!-- ■■ TESTIMONIALS ■■ -->
<section class="py-5 bg-primary text-white">
    <div class="container">
        <h2 class="text-center fw-bold mb-5">What Our Students Say</h2>
        <div class="row g-4">
            <div class="col-md-4">
                <div class="card bg-white bg-opacity-10 border-0 text-white h-100 p-4">
                    <p class="fst-italic">"The best web development course in Karad.
                    Ranjiit Sir explains every concept so clearly with practical examples."</p>
                    <div class="d-flex align-items-center mt-3 gap-3">
                        <div class="rounded-circle bg-warning d-flex align-items-center
                        justify-content-center" style="width:42px;height:42px;">
                            <strong class="text-dark">A</strong>
                        </div>
                    </div>
                </div>
            </div>
        </div>
    </div>

```

```

        <div class="fw-bold">Aarav Sharma</div>
        <small class="opacity-75">BCA Graduate, Karad</small>
      </div>
    </div>
  </div>
</div>
<div class="col-md-4">
  <div class="card bg-white bg-opacity-10 border-0 text-white h-100 p-4">
    <p class="fst-italic">"Got placed as a frontend developer within 2 months
      of completing the course. The projects really helped my portfolio."</p>
    <div class="d-flex align-items-center mt-3 gap-3">
      <div class="rounded-circle bg-success d-flex align-items-center
        justify-content-center" style="width:42px;height:42px;">
        <strong class="text-white">P</strong>
      </div>
      <div>
        <div class="fw-bold">Priya Patil</div>
        <small class="opacity-75">Frontend Developer, Pune</small>
      </div>
    </div>
  </div>
</div>
<div class="col-md-4">
  <div class="card bg-white bg-opacity-10 border-0 text-white h-100 p-4">
    <p class="fst-italic">"Excellent study materials! The PDF notes are so
      detailed and the homework problems really tested my understanding."</p>
    <div class="d-flex align-items-center mt-3 gap-3">
      <div class="rounded-circle bg-danger d-flex align-items-center
        justify-content-center" style="width:42px;height:42px;">
        <strong class="text-white">R</strong>
      </div>
      <div>
        <div class="fw-bold">Rohan Desai</div>
        <small class="opacity-75">B.Sc CS Student</small>
      </div>
    </div>
  </div>
</div>
</div>
</div>
</section>

<!-- ■■ CONTACT ■■ -->
<section class="py-5" id="contact">
  <div class="container">
    <h2 class="text-center fw-bold section-title mb-5">Get In Touch</h2>
    <div class="row g-5 justify-content-center">
      <div class="col-lg-5">
        <h3 class="fw-bold mb-4">Contact Information</h3>
        <div class="d-flex gap-3 mb-3 align-items-start">
          <span class="fs-4">&#128205;</span>
        </div>
      </div>
    </div>
  </div>
</section>

```

```

        <strong>Address</strong><br>
        <span class="text-muted">ForkInfosystems, Karad, Maharashtra – 415110</span>
    </div>
</div>
<div class="d-flex gap-3 mb-3 align-items-start">
    <span class="fs-4">#128241;</span>
    <div>
        <strong>Phone</strong><br>
        <a href="tel:9172135355" class="text-decoration-none text-primary">
            9172135355</a> /
        <a href="tel:7757962804" class="text-decoration-none text-primary">
            7757962804</a>
    </div>
</div>
<div class="d-flex gap-3 mb-4 align-items-start">
    <span class="fs-4">#128231;</span>
    <div>
        <strong>Email</strong><br>
        <a href="mailto:info@forksystems.com" class="text-decoration-none text-primary">
            info@forksystems.com</a>
    </div>
</div>
<div class="alert alert-info">
    <strong>#128337; Batch Timings:</strong><br>
    Morning: 9 AM – 11 AM | Evening: 6 PM – 8 PM<br>
    Monday to Saturday
</div>
</div>
<div class="col-lg-6">
    <div class="card border-0 shadow-sm p-4">
        <h5 class="fw-bold mb-4">Enquire Now</h5>
        <form>
            <div class="row g-3">
                <div class="col-sm-6">
                    <div class="form-floating">
                        <input type="text" class="form-control" id="cName"
                            placeholder="Name" required>
                        <label for="cName">Full Name *</label>
                    </div>
                </div>
                <div class="col-sm-6">
                    <div class="form-floating">
                        <input type="tel" class="form-control" id="cPhone"
                            placeholder="Phone">
                        <label for="cPhone">Phone Number</label>
                    </div>
                </div>
                <div class="col-12">
                    <div class="form-floating">
                        <input type="email" class="form-control" id="cEmail"
                            placeholder="Email" required>
                    </div>
                </div>
            </div>
        </form>
    </div>
</div>

```



```
    </div>
</footer>

<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.js"
>
</script>
</body>
</html>
```

Homework — Module-wise Questions

Module 1 — Introduction

[T Theory Questions]

T1. What is Bootstrap? Why was it created?

Ans: _____

T2. What is the difference between Bootstrap CDN and downloading Bootstrap locally?

Ans: _____

T3. Why is the viewport meta tag required for Bootstrap?

Ans: _____

T4. What is the JS bundle and when is it required?

Ans: _____

T5. How do Bootstrap utility classes differ from writing custom CSS?

Ans: _____

[P Practical Exercises]

P1. Create a basic Bootstrap page with a heading, paragraph, and two buttons (primary and danger).

Ans: _____

P2. Build a page that uses text-primary, text-success, bg-warning, and bg-dark.

Ans: _____

P3. Create a page using Bootstrap to display your institute information.

Ans: _____

Module 2 — Grid System

[T Theory Questions]

T1. Explain the Bootstrap grid system. What are the three required elements?

Ans: _____

T2. What is a breakpoint in Bootstrap? List all 6 breakpoints.

Ans: _____

T3. What is the difference between container and container-fluid?

Ans: _____

T4. How do you offset a column? Give an example.

Ans: _____

T5. Explain row gutters. What is the difference between g-3, gx-3, and gy-3?

Ans: _____

[P Practical Exercises]

P1. Create a 3-equal-column layout that stacks on mobile using col-md-4.

Ans: _____

P2. Build a sidebar (col-lg-3) + main content (col-lg-9) layout.

Ans: _____

P3. Create a responsive photo grid: 1 col on mobile, 2 on tablet, 4 on desktop.

Ans: _____

Module 3 & 4 — Typography & Spacing

[T Theory Questions]

T1. List 5 Bootstrap typography classes and what each does.

Ans: _____

T2. Explain the naming pattern for Bootstrap spacing (margin/padding) classes.

Ans: _____

T3. What is mx-auto used for? Give an example of when you would use it.

Ans: _____

T4. What is the difference between d-none and invisible?

Ans: _____

[P Practical Exercises]

P1. Build a page using display-1 through display-6 headings.

Ans: _____

P2. Style a blockquote with author citation using Bootstrap classes.

Ans: _____

P3. Create a layout with specific spacing: m-4 on card, px-5 on text, gap-3 on button row.

Ans: _____

Module 5 — Flexbox

[T Theory Questions]

T1. What class makes an element a flex container in Bootstrap?

Ans: _____

T2. How do you justify content to space-between using Bootstrap?

Ans: _____

T3. What does ms-auto do in a flex row? Give a navbar example.

Ans: _____

T4. How do you create a vertically and horizontally centred layout using Bootstrap flex?

Ans: _____

[P Practical Exercises]

P1. Build a horizontal navbar with logo on left, links in centre, button on right using d-flex.

Ans: _____

P2. Create a card row with 3 cards that wrap on small screens using d-flex flex-wrap.

Ans: _____

P3. Build a footer with two columns: info on left, links on right using d-flex justify-content-between.

Ans: _____

Module 6 — Buttons & Alerts

[T Theory Questions]

T1. List all Bootstrap button colour variants (8 solid, 8 outline).

Ans: _____

T2. How do you create a full-width button?

Ans: _____

T3. What is the difference between a badge and an alert?

Ans: _____

T4. How do you make an alert dismissible?

Ans: _____

[P Practical Exercises]

P1. Build a button toolbar with 5 different button styles and sizes.

Ans: _____

P2. Create a notification area with success, warning, and danger alerts (one dismissible).

Ans: _____

P3. Add badges to a nav menu showing notification counts.

Ans: _____

Module 7 — Navbar & Navigation

[T Theory Questions]

T1. What does navbar-expand-lg mean? What happens on smaller screens?

Ans: _____

T2. How do you add a dropdown menu inside a navbar?

Ans: _____

T3. What is the difference between nav-tabs and nav-pills?

Ans: _____

T4. How do you make a navbar sticky at the top of the page?

Ans: _____

[P Practical Exercises]

P1. Build a complete navbar with: logo, 4 nav links, a dropdown, and a search bar.

Ans: _____

P2. Create a tab component with 3 tabs (HTML, CSS, JS) showing different content.

Ans: _____

P3. Build a breadcrumb + pagination component for a blog page.

Ans: _____

Module 8 — Cards & Images

[T Theory Questions]

T1. What is the basic structure of a Bootstrap card? List all card sub-classes.

Ans: _____

T2. How do you make all cards in a row equal height?

Ans: _____

T3. What is img-fluid? When should you use it?

Ans: _____

T4. What does the ratio class do? Give an example for 16:9 YouTube embed.

Ans: _____

[P Practical Exercises]

P1. Create a course catalog: 3 cards in a responsive grid, each with image, title, price, button.

Ans: _____

P2. Build a team section: horizontal cards with image on left, info on right (col-md-4 + col-md-8).

Ans: _____

P3. Create a gallery with 6 images in a responsive grid with img-thumbnail styling.

Ans: _____

Module 9 — Forms

[T Theory Questions]

T1. What class do you apply to an input to make it Bootstrap-styled?

Ans: _____

T2. What is form-floating? How does it work?

Ans: _____

T3. What is an input group? Give 2 examples.

Ans: _____

T4. How does Bootstrap form validation work? What JS is needed?

Ans: _____

[P Practical Exercises]

P1. Build a complete Registration Form with floating labels, all input types, and validation.

Ans: _____

P2. Create a search bar using input-group with a dropdown category selector and search button.

Ans: _____

P3. Build a multi-column form layout using the Bootstrap grid inside a form.

Ans: _____

Module 10 & 11 — Components & JS Plugins

[T Theory Questions]

T1. What is a Modal? Write the minimum HTML needed for a working modal.

Ans: _____

T2. How do you trigger a Bootstrap component using data attributes vs JavaScript API?

Ans: _____

T3. What is an Offcanvas? How does it differ from a Modal?

Ans: _____

T4. What is Scrollspy and when is it used?

Ans: _____

T5. How do you initialize tooltips on a page using JavaScript?

Ans: _____

[P Practical Exercises]

P1. Build a confirm-delete modal: button triggers modal, modal has cancel + confirm buttons.

Ans: _____

P2. Create an FAQ page using Bootstrap Accordion with 5 questions.

Ans: _____

P3. Add a carousel with 3 slides (use placeholder images) with captions, indicators, and controls.

Ans: _____

P4. Create a sidebar navigation using Offcanvas that opens from the left side.

Ans: _____